

CS560 Midterm #1

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I. QUESTION (12 POINTS)

Consider these documents:

Doc1: new home sales top forecasts

Doc2: home sales rise in july

Doc3: increase in home sales in july

Doc4: july new home sales rise

- 1) Draw the term-document incidence matrix for this document collection.
- 2) Draw the inverted index representation for this collection.

II. QUESTION (12 POINTS)

Write the algorithm that handles the boolean query $x \text{ XOR } y$ using inverted index.

III. QUESTION (12 POINTS)

Apply filters

- Case-folding
- Stemmer
- Stop word removal

to the document

Such an analysis can reveal features that are not easily visible from the Variations in the individual genes and can lead to a Picture of expression that is more biologically Transparent and accessible to interpretation

IV. QUESTION (10 POINTS)

Create a portion of a positional index such that the first query “fools rush in” returns documents with id’s 4, 7, 10, 14; the second query “fools rush in” AND “fools fear” returns documents with id’s 4, 14; and the third query “fools fear” returns documents with id’s 1, 4, 12, 14.

V. QUESTION (12 POINTS)

Give two words of length 5 whose edit distance is 3. Compute their edit distance using dynamic programming.

VI. QUESTION (12 POINTS)

Write the pseudocode for front coding of N tokens. For example, if we compress the words

```

automata
automate
automatic
automation

```

using front coding, we get

```
8automat$al#e2#ic#3#ion
```

VII. QUESTION (10 POINTS)

If there are 4 zones in a document and the weights of these zones are $g_0 = 0.1$, $g_1 = 0.2$, $g_2 = 0.3$ and $g_3 = 0.4$, what are all the distinct score values a document may get?

VIII. QUESTION (12 POINTS)

Let say we use nnn (direct term frequency, 1 as document frequency and no normalization) as tf-idf variant in query ranking. Construct a query such that the resulting ranking will be 1, 3, 2, 4 for the documents in Question 1.

IX. QUESTION (8 POINTS)

An IR system returns 8 relevant documents, and 10 non-relevant documents. There are a total of 20 relevant documents in the collection. What is the precision of the system on this search, and what is its recall?